

Zomato Restaurant Insights Dashboard

Case Study Documentation

Client: Head of Growth & Partnerships, a restaurant-discovery and food-delivery platform (client role-played)

Analyst role: Data Analyst

Reference: [GeeksforGeeks - Zomato Data Analysis Using Python](#)

1. Business Background

We list thousands of restaurants and earn revenue from online orders, table bookings and partner subscriptions. Leadership wants to understand how restaurants in our catalogue perform, what customers prefer, and where we should focus on-boarding and marketing effort. Today this analysis is done ad hoc in spreadsheets, which is slow and inconsistent. We want a repeatable analysis and a self-serve dashboard.

2. Problem Statements

1. **Online adoption gap:** We do not know how many listed restaurants accept online orders versus offline-only, and whether this differs by restaurant type.
2. **Category demand:** We cannot say which restaurant categories (Dining, Cafes, Buffet, etc.) attract the most customer engagement, measured by votes.
3. **Pricing sweet spot:** We lack clarity on what price range couples prefer for dining out, which affects promotions and deal design.
4. **Quality perception:** We do not know how ratings are distributed, or whether online-enabled restaurants are rated differently from offline-only ones.
5. **Star performers:** We need a quick way to identify the most-voted restaurants for featured placements and partnership outreach.
6. **No self-serve reporting:** Business teams depend on analysts for every question and want a single dashboard with filters.

3. Objectives

- Quantify online vs. offline order availability, overall and by restaurant type.
- Identify the most popular restaurant categories and top-voted restaurants.
- Identify the preferred cost range for two people.
- Understand the ratings distribution and its relationship with online ordering.
- Deliver an interactive dashboard plus a short findings summary with recommendations.

4. Dataset

Item	Detail
Download link	https://media.geeksforgeeks.org/wp-content/uploads/20250117023324808265/Zomato-data-.csv
Format	CSV
Source article	geeksforgeeks.org/data-analysis/zomato-data-analysis-using-python/
Solution notebook	Zomato_Data_Analysis_Using_Python.ipynb

Fields used in the analysis

Column	Description
name	Restaurant name
online_order	Whether the restaurant accepts online orders (Yes/No)
rate	Rating stored as text such as "4.1/5" (needs cleaning to a number)
votes	Number of customer votes
approx_cost(for two people)	Approximate cost of a meal for two, in INR
listed_in(type)	Restaurant category (Dining, Cafes, etc.)

Data preparation needs

- Convert **rate** from text ("4.1/5") to a float.
- Confirm there are no nulls (the tutorial reports none) and check for duplicate restaurant names.
- Ensure the cost column is numeric and consistently formatted.

Caveat: The dataset is a small sample, so findings are indicative, not representative of the whole market.

5. Questions the Client Will Ask

A. Online vs. Offline

1. How many restaurants accept online orders, and how many do not? What is the percentage split?
2. Which restaurant types have the highest and lowest share of online ordering?
3. Are cafes really more online-driven than dining restaurants?

B. Restaurant Types and Popularity

4. Which restaurant type has the most listings?
5. Which type collects the most total votes?
6. Is a type popular because it has many restaurants, or because each restaurant gets many votes? (average votes per restaurant by type)
7. Which restaurant has the highest number of votes overall, and what are the top 10?

C. Ratings

8. What does the ratings distribution look like? What are the average and median ratings?
9. What percentage of restaurants are rated 4.0 and above, and below 3.0?
10. Do online-order restaurants have higher ratings than offline-only ones?
11. Do higher-rated restaurants also get more votes?

D. Pricing

12. What is the most common cost for two people?
13. Which price band (under Rs. 300, Rs. 300-500, Rs. 500-800, above Rs. 800) has the most restaurants?
14. How does cost relate to rating? Do pricier restaurants score better?
15. Which price bands are most common within each restaurant type?

E. Strategy and Action

16. Where should we focus on-boarding to increase online-order coverage?
17. Which restaurants should we feature in campaigns?
18. Which price range should our couple-dining deals target?
19. Are there under-rated restaurants (high votes, low rating) that need quality follow-up?

6. Dashboard Requirements

6.1 Purpose and Audience

A one-stop, interactive dashboard for the Growth, Partnerships and Marketing teams, readable in under two minutes by a non-technical user.

6.2 Tool

Power BI or Tableau preferred (Streamlit or Looker Studio acceptable). Data refresh from the cleaned CSV at minimum.

6.3 Layout: three pages

Page 1: Overview

Component	Type	Answers
Total restaurants, average rating, total votes, % online, median cost for two	KPI cards	-
Online vs. offline split	Donut chart	Q1
Restaurants by type	Bar chart	Q4
Rating distribution	Histogram	Q8, Q9

Page 2: Popularity and Ratings

Component	Type	Answers
Votes by restaurant type (total / average toggle)	Bar chart	Q5, Q6
Top 10 restaurants by votes	Ranked bar / table	Q7, Q17
Rating by online-order status	Box plot	Q10
Votes vs. rating	Scatter plot	Q11, Q19
Type vs. online order	Heatmap with counts	Q2, Q3

Page 3: Pricing

Component	Type	Answers
Distribution of cost for two	Column chart	Q12
Restaurants per price band	Bar chart	Q13
Cost vs. rating	Scatter plot	Q14
Price band by restaurant type	Stacked bar	Q15

6.4 Filters and Interactivity

- Global slicers: restaurant type, online order (Yes/No), rating range, cost range.
- Cross-filtering between charts; hover tooltips with name, rating, votes and cost.
- Drill-through from a restaurant type to its restaurant list; a reset-filters button.
- Table view exportable to CSV/Excel.

6.5 Design Guidelines

- Clean layout with a consistent palette (a red accent aligned to our brand is welcome).
- Every chart carries a clear title and axis labels; cost is labelled in INR.
- A short "Key takeaway" caption under each major chart.

- Works on desktop and tablet; the Overview page should be mobile-friendly.

6.6 Data and Quality Requirements

- Cleaned ratings stored as numeric values.
- A documented data dictionary and transformation steps.
- A visible "data as of" date and a note about sample size.

6.7 Non-Functional Requirements

- Page load under 5 seconds; view-only role-based access for business users.
- Manual or scheduled refresh when new data arrives.

7. Deliverables

1. Cleaned dataset and a data dictionary.
2. Python notebook with the exploratory analysis (pandas, Matplotlib, Seaborn).
3. Interactive dashboard (3 pages as above).
4. A 2-page findings summary with recommendations.
5. A short walkthrough session for stakeholders.

8. Success Criteria

- All 19 questions in Section 5 can be answered from the dashboard without analyst help.
- Dashboard numbers match the notebook output.
- Business users can find the top restaurants, online-order split and preferred price band within a couple of clicks.

9. Assumptions and Risks

Item	Detail
Sample size	The data is a limited sample, so conclusions are directional.
Correlation vs. causation	Higher ratings among online-enabled restaurants may reflect restaurant type or price, not the online option itself.
Snapshot data	No time dimension, so trend analysis is out of scope.
Currency	All costs are assumed to be in INR.

10. Suggested Timeline

Phase	Duration
Data cleaning and exploration	2 days
Analysis and insights	2 days
Dashboard build	3 days
Review, revisions and handover	2 days